

JIAHE JIANG

236-984-0398 ◇ Vancouver, BC

jiahe72@student.ubc.ca ◇ github.com/JiaheJiang ◇ linkedin.com/in/jiahecrystaljiang

OBJECTIVE

Fourth-year Computer Science student with experience in healthcare technology, data analytics, AI-assisted learning tools, and research software development. Skilled in Python, SQL, Power BI, machine learning workflows, dashboard development, technical documentation, and cross-functional communication.

EDUCATION

Bachelor of Science, Major in Computer Science

09/2022 - Expected 2026

University of British Columbia - Vancouver, BC

Year 4

Cumulative GPA: 4.10/4.33

Relevant Coursework: Programming Languages, Algorithms and Data Structures, Software Construction, Computer Systems, Applied Machine Learning, Data Mining, Data Analysis.

SKILLS

Programming	Python, Java, C++, C, SQL, R, JavaScript, HTML, CSS, React
Machine Learning	PyTorch, TensorFlow, scikit-learn, NumPy, pandas, feature engineering
Data & Analytics	Power BI, Tableau, Streamlit, data preprocessing
Software	VS Code, Jupyter Notebook, Git/GitHub, Conda, RStudio, IntelliJ, Figma, MS Office
Testing	JUnit, GDB, Makefile

WORK EXPERIENCE

Technology & Innovation Assistant

09/2025 - 04/2026

UBC Faculty of Education - Learning Design & Digital Innovation

Vancouver, BC

- Supported the development of UBC PointQuest, an interactive learning platform, by contributing to role-based access control and secure permission management for instructor and student users.
- Designed and delivered AI literacy workshops on LLM-based tools, AI-assisted presentations, voice synthesis, and responsible AI use for students, faculty, and staff.
- Collaborated with faculty, learning designers, and technical staff to translate academic needs into practical digital learning solutions.

Digital Information Technologies & Solutions Co-op

09/2024 - 08/2025

Sanofi

Toronto, ON

- Delivered cross-departmental technical support for enterprise manufacturing systems, including MES and PI, helping maintain operational continuity in a regulated pharmaceutical environment.
- Performed GxP-compliant qualification and validation for 100+ pharmaceutical peripherals, ensuring audit-ready documentation and regulatory adherence.
- Built and maintained Power BI activity-monitoring dashboards to support cybersecurity tracking, user-activity visibility, and data-exposure risk reduction.

R&D Assistant Engineer

06/2023 - 07/2023

Wuhan Deepin Technology Company

Wuhan, China

- Coordinated R&D project timelines, deliverables, and technical documentation to support product development planning and milestone tracking.



- Evaluated emerging tools, techniques, and engineering workflows to improve development efficiency and assess feasibility for future software projects.
- Assisted with control design, implementation, testing, and integration activities within an agile engineering team.

RESEARCH EXPERIENCE

Research Assistant

01/2026 - Present

UBC Department of Electrical and Computer Engineering - Lele's Lab

Hybrid

- Developed an interactive Streamlit dashboard for power-grid vulnerability analysis, integrating graph visualization, topology editing, metric panels, attack-strategy filters, and session recording.
- Refactored Jupyter Notebook prototypes into modular Python scripts by separating simulation, visualization, and robustness-analysis logic for cleaner execution and future dashboard integration.

Research Assistant

01/2026 - 04/2026

UBC Department of Chemistry - Huan Lab

On-Site

- Developed Python-based mass spectrometry preprocessing workflows for blank-sample alignment, peak filtering, and feature-level comparison to support contaminant and unknown-compound discovery.
- Adapted MS/MS analysis workflows for iodinated compound prediction, downstream result cleaning, and interpretation of large nontargeted datasets.
- Built reference-library workflows to support spectral matching, compound annotation, and future contaminant screening pipelines.

Research Assistant

09/2024 - 04/2026

Huazhong University of Science and Technology

Remote

- Implemented machine learning research pipelines for graph learning and multi-view clustering experiments, including model training, hyperparameter tuning, and result evaluation.
- Supported communication-signal processing projects by implementing OFDM simulation modules, tuning Doppler-resilience parameters, and evaluating model performance.
- Contributed to a model-driven deep learning detector for AFDM systems by helping implement trainable neural modules for improved symbol detection and convergence stability.

PROJECTS

Machine Learning and Data Processing Workflows. Built ML-ready preprocessing pipelines for scientific datasets, including feature cleaning, assignment validation, matrix construction, and prediction-output analysis using Python, pandas, NumPy, and scikit-learn.

Interactive Analytics Dashboard. Developed a Streamlit dashboard for graph-based infrastructure analysis, integrating visualization, user controls, metric panels, attack-strategy filters, and experiment-state tracking.

Cosmetic Warehouse App. Constructed a Java desktop application with a JavaFX GUI to manage warehouse inventory. Implemented backend data tracking and local persistence using JSON storage.

Student Success & Parental Education. Analyzed GPA and parental education data using R, bar plots, histograms, hypothesis testing, confidence intervals, and model-performance evaluation.

HONORS & LANGUAGES

- **Awards:** Work Learn International Undergraduate Research Awards; Dean's List; Outstanding International Student Award; Go Global International Learning Programs Award.
- **Languages:** Mandarin (Native or Bilingual); English (Professional Working).